TEXORA · COURSE NOTES · LECTURE SHEET

Fluid Mechanics

Fluids at rest and in motion — pressure in a liquid, Pascal's and Archimedes' principles, streamline flow, the continuity equation, Bernoulli's theorem and its instruments.

PHY-1103 · PHYSICS-I · TOPIC 4 · LEVEL 1 TERM 1

1. Fluids at rest: pressure and two principles

A fluid — liquid or gas — cannot sustain a shear stress (Topic 2: its rigidity modulus is zero), so at rest it exerts only **pressure**: a normal force per unit area, equal in all directions at a point. In a liquid of density ρ the pressure grows with depth because each layer carries the weight of those above:

$$p = p_0 + \rho\, g\, h$$

with p_0 the pressure on the surface and h the depth. Two classical principles organise everything else in hydrostatics. **Pascal's law:** pressure applied to an enclosed fluid is transmitted undiminished to every part of it — the working secret of the hydraulic press and brake, where a small force on a small piston reappears multiplied by the piston-area ratio on a large one. **Archimedes' principle:** a body immersed in fluid is buoyed up by a force equal to the weight of the fluid it displaces, $F = \rho_{fluid}\, V\, g$ — the reason ships float, hydrometers read densities, and a submerged body's apparent weight loss exactly equals the displaced water's weight. A body floats when its average density is below the fluid's, displacing its own weight of fluid before it is fully under.

2. Fluids in motion: streamlines and Reynolds number

In **streamline (laminar) flow** every particle follows the smooth path of the particle before it; the flow is layered and quiet. Speed it up, or widen the channel, and the flow breaks into eddies — **turbulent flow**. The borderline is predicted by the dimensionless **Reynolds number**,

$$Re = \rho\, v\, D / \eta$$

(v speed, D a characteristic size, η viscosity): for flow in a pipe, laminar below about 2000, turbulent above about 4000, unstable between. Reynolds number is the engineer's traffic light for every flow in this course — including the flow of dye liquor through a fabric package, where laminar flow penetrates evenly and turbulence is engineered deliberately in jet machines. Viscosity itself is Topic 5; here it simply enters as the flow's internal friction.

3. The equation of continuity

For an incompressible fluid in steady flow, what flows in must flow out: the mass passing each cross-section per second is constant, ρAv = const, and with ρ constant,

$$A_1 v_1 = A_2 v_2$$

Narrow the pipe and the fluid speeds up in exact inverse proportion to the area. A garden hose pinched at the end, blood speeding through a narrowed artery, liquor accelerating through the constrictions of a yarn package — one equation. Worked: a pipe of cross-section 4.0 cm² carrying water at 0.50 m s⁻¹ narrows to 1.0 cm²; the speed there is $v_2 = 0.50 \times 4.0/1.0 =$ **2.0 m s⁻¹**.

4. Bernoulli’s theorem

For steady, incompressible, non-viscous flow, applying the work–energy theorem along a streamline gives **Bernoulli’s theorem**: the quantity

$$p + \tfrac{1}{2} \rho v^2 + \rho g h = \text{constant (along a streamline)}$$

— pressure energy, kinetic energy and potential energy per unit volume trade among themselves but sum to a constant. Its most used consequence: where a fluid speeds up, its pressure falls. The theorem’s family of instruments and effects:

- **Venturi meter:** a constricted section where the fluid speeds up and the pressure drop, read on a manometer, measures the flow rate — Bernoulli plus continuity, calibrated.
- **Torricelli’s law:** liquid escaping from a hole at depth h below the surface of a large tank emerges at $v = \sqrt{(2gh)}$ — free-fall speed from that height, as if each particle had simply fallen.
- **Atomisers and sprayers:** a fast air jet over a tube’s mouth drops the pressure there and lifts liquid into the airstream — the paint spray, the perfume spray, and the spray application of textile finishes.
- **Lift and curves:** air faster over a wing’s curved top means lower pressure above (a simplified reading of lift), and a spinning ball’s curved path (Magnus effect) is the same physics in sport.

WORKED EXAMPLE — BERNOULLI WITH CONTINUITY

Water flows through the venturi of §3: wide section $A_1 = 4.0$ cm², $v_1 = 0.50$ m s⁻¹, narrow section $A_2 = 1.0$ cm², $v_2 = 2.0$ m s⁻¹, both at the same height. Pressure drop: $p_1 - p_2 = \tfrac{1}{2}\rho(v_2^2 - v_1^2) = \tfrac{1}{2} \times 1000 \times (4.0 - 0.25) =$ **1875 Pa** — about 19 cm of water column on a manometer. That single measurement of Δp, with the areas known, returns the flow rate $Q = A_1 v_1 = 2.0 \times 10^{-4}$ m³ s⁻¹.

The theorem’s fine print matters: real (viscous) fluids lose energy to friction along the flow, so Bernoulli balances only approximately over long pipes — the correction is the engineer’s head-loss

term, and the honesty is Topic 5's.

5. The textile thread

Wet processing is fluid mechanics with product at stake. In **package dyeing**, liquor is pumped through a wound yarn package — a porous plug — and the flow pattern decides levelness: channelling (fluid finding easy paths, a continuity-and-resistance problem) dyes unevenly. In **jet dyeing**, a venturi jet of liquor transports delicate fabric gently — Bernoulli's pressure drop doing the work, with the jet's Reynolds number high and turbulent by design. **Air-jet spinning and weaving** replace water with air jets that draft, twist and insert weft at extreme speed. Spray finishing, foam application, the hydroextractor's drainage, even the hydrometer checking a bath's density on Archimedes' principle — the dyehouse runs on this topic from end to end.

6. Glossary — key terms of this topic

TERM	MEANING
Pressure	Normal force per unit area in a fluid; at depth h in a liquid, $p = p_0 + \rho gh$.
Pascal's law	Pressure applied to an enclosed fluid is transmitted equally throughout it.
Archimedes' principle	Upthrust on an immersed body equals the weight of fluid displaced.
Streamline flow	Smooth, layered flow in which particles follow fixed paths; contrasted with turbulent flow.
Reynolds number	$Re = \rho vD/\eta$; predicts the laminar–turbulent transition (≈2000/4000 in pipes).
Continuity equation	$A_1v_1 = A_2v_2$ for steady incompressible flow.
Bernoulli's theorem	$p + \frac{1}{2}\rho v^2 + \rho gh$ is constant along a streamline for ideal steady flow.
Venturi meter	A constricted pipe section whose pressure drop measures flow rate.

7. Key points and summary

- Hydrostatics: $p = p_0 + \rho gh$, Pascal's transmission of pressure, Archimedes' upthrust = weight of fluid displaced.
- Flow regimes are set by Reynolds number; continuity ($A_1v_1 = A_2v_2$) makes fluids speed up through constrictions.
- Bernoulli trades pressure, speed and height along a streamline — explaining venturi meters, Torricelli efflux, atomisers and the jet machinery of the dyehouse.

8. Practice

Answer in your own words or with full working; brief model answers follow.

1. Find the pressure due to the water alone at a depth of 5.0 m ($\rho = 1000$ kg m^{-3}), and the total pressure there including the atmosphere (1.01×10^5 Pa).
2. A hydraulic press has pistons of area 5 cm² and 500 cm². A force of 40 N is applied to the small piston. Find the force at the large piston and name the law at work.
3. A block of volume 0.002 m³ is fully immersed in water. Find the upthrust on it.
4. Liquor flows at 1.2 m s^{-1} in a pipe of 6.0 cm² cross-section, then enters a section of 2.0 cm². Find its new speed.
5. Find the speed at which water escapes from a hole 0.80 m below the surface of an open tank (Torricelli).
6. For flow of a liquor ($\rho = 1050$ kg m^{-3}, $\eta = 2.0 \times 10^{-3}$ Pa s) at 1.5 m s^{-1} through a pipe of diameter 0.050 m, compute Re and state the flow regime.
7. Explain how a venturi meter turns a pressure reading into a flow rate, naming the two equations used.
8. In package dyeing, what goes wrong if the liquor channels through a few easy paths instead of flowing uniformly — explain in the language of this topic.

Answer key

1. $\rho gh = 1000 \times 9.8 \times 5.0 = 4.9 \times 10^4$ Pa; total $= 1.01 \times 10^5 + 0.49 \times 10^5 = 1.50 \times 10^5$ Pa.
2. Pressure is transmitted equally: $F_2 = F_1 \times A_2/A_1 = 40 \times 500/5 = 4000$ N. Pascal’s law.
3. Upthrust $= \rho Vg = 1000 \times 0.002 \times 9.8 = 19.6$ N.
4. $v_2 = 1.2 \times 6.0/2.0 = 3.6$ m s^{-1}.
5. $v = \sqrt{(2gh)} = \sqrt{(2 \times 9.8 \times 0.80)} = \sqrt{15.68} \approx 3.96$ m s^{-1}.
6. $Re = 1050 \times 1.5 \times 0.050 / 0.002 = 39{,}375 \approx 3.9 \times 10^4$ — well above 4000, so fully turbulent.
7. Continuity gives the speed increase at the throat from the area ratio; Bernoulli converts the speed increase into a pressure drop; measuring Δp (manometer) therefore yields v and hence $Q = Av$.
8. Continuity forces the total flow through whatever paths exist; easy paths take high local speed while dense regions are starved, so those regions receive less dye per unit time — the package dyes unevenly (channelled, unlevel result).